

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

19th November 2024

CP1A - Actuarial Practice

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

- Q. 1)** a) Outline *the two* approaches to financing pension benefits. (2)
- b) Briefly explain the various methods of funding the benefits. (5)
- [7]
- Q. 2)** A life insurance company LifeInvest Ltd. is considering expanding into the general insurance industry by establishing a new company General Co Ltd.
- a) Briefly mention how the following aspects of general insurance differ from that of a life insurance:
- Nature of risks
 - Policy duration
 - Frequency of claims
 - Reserve requirements
- (4)
- b) How does the investment strategy and asset portfolio of a general insurance company differ from that of a life insurance company in view of the differences mentioned in part (a)? (4)
- c) Gold prices in the last 1 year have increased by 40%. General Co. wants to invest 75% of its liquid assets in physical gold instead of money markets. You have been hired as an actuarial consultant to advice on this strategy. Outline the points you would highlight regarding this strategy in your report to the board of directors. (6)
- [14]
- Q. 3)** A life insurance company sells only term insurance products.
- a) Discuss the various underwriting aspects that need to be considered to assess the risk of a life. (4)
- b) Describe the risk management tools the insurance company can use to manage and control the risks from this product. (8)
- [12]
- Q. 4)** There is a large bridge through which major trade of the world passes through. Due to an unfortunate accident, a ship has collided into it and as a result, the bridge has collapsed.
- The following statement has been made by a major news outlet “*Disputes over liability and the cost of claims could take years for insurers to resolve and is expected to result in billions of dollars in payouts for the insurers. The critical aspect of the settlement of dispute is to fix the responsibility of various parties involved. This process is expected to be expensive, complex and lengthy.*”
- a) Comment and analyse the statement reported in the news outlet including the potential types of insurance classes/lines of business this incident could impact. Your answer should cover the following areas:
- Classes / Lines of business
 - Legal issues
 - Reinsurance
- (14)

- b) Analyse the potential impact on rates for marine insurance and reinsurance because of this incident.

(3)
[17]

Q. 5) The insurance market in the country of Utopia is a fast-growing market with current insurance penetration being low. The insurance regulator of this country wants to increase insurance penetration and is exploring ways of promoting innovation to do so.

- a) Describe possible ways in which the regulator can promote innovation in the insurance industry.

(5)

The current motor insurance market in Utopia is largely driven by agents and brokers.. A new digital company DigitalCo. sells motor and health insurance through digital channels directly to the customer. DigitalCo. At present the company has a combined ratio of 170%. As part of promoting innovation, the regulator has removed the controls around pricing and policy wordings of motor insurance policies.

- b) Explain the impact of regulator's action of decontrolling the motor insurance business pricing on DigitalCo.

(8)

A private equity investor is considering to invest in Digital Co.

- c) Outline why would an investor consider funding DigitalCo. despite it having a high combined ratio.

(5)
[18]

Q. 6) An established company operating a chain of hospitals in the country of Actuaria has been granted a license to start a standalone health insurance company. You have been appointed as a consulting actuary for the company.

- a) Describe the **key aspects and risks** the hospital chain has considered before applying to the regulator for a health insurance license.

(9)

- b) List the policy and claims data fields that the new company should capture.

(3)

Since the company is new and doesn't have data, the appointed actuary of the company is considering using industry data to price new products.

- c) Describe the issues with using industry data and ways to address such issues.

(4)
[16]

Q. 7) The central bank of an economy has raised repo rates due to continued high rates of inflation of 9% p.a. each in the last 4 quarters. The long-term bond portfolio of a general insurer writing comprehensive lines of business has faced notional losses following the rate increase.

- a) Briefly explain the reasons why a general insurer would invest in long term bonds?

(2)

The insurer is considering raising capital by way of subordinated debt at 10%p.a interest to meet its short-term capital and payout needs. A reinsurer has approached the insurer with an offer for writing a quota share business that would address the capital needs of the insurer. The insurer has a loss ratio of 75% and the proposed fixed commission is 20%. The reinsurer will have a margin to cover its profit and expenses. The reinsurer calculates its margin as follows:

Margin = 1 – Loss Ratio – Commission

The reinsurer has stated “In the current economic environment, reinsurance is cheaper than subordinated debt”

b) Outline how a quota share can help the insurer with its capital needs? (3)

c) Evaluate the reinsurer’s statement. (4)

The insurance company has decided to enter into a quota share arrangement with the reinsurer. The reinsurer and insurer have a different view on the loss ratio and hence have decided on a profit commission arrangement as part of the quota share agreement.

d) Define profit commission. (2)

e) Describe the other capital management tools this insurer could use and how it would address the needs of the insurer. (5)

[16]
